



AI-Powered Interview Preparation System: Integrating Resume Analysis, HR Simulation, and Technical Skill Assessment

**Harsh Koshti ^{a++*}, Prathamesh Gosavi ^{a++}, Roshan Pagar ^{a++},
Prathamesh Khairnar ^{a++} and Sopan Talekar ^{a#}**

^a *MVPS KBT College of Engineering, Nashik, Maharashtra, India.*

Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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ABSTRACT

This article presents an AI-powered interview preparation system designed to objectively collect and evaluate candidate responses. The solution automates key stages of the interview process, enhancing recruitment efficiency through real-time emotion detection using Convolutional Neural Networks (CNN), natural language processing (NLP) for resume analysis, and speech recognition for response evaluation. The system integrates with an Applicant Tracking System (ATS) for resume screening, an HR interview simulator focusing on communication and behavioural skills, and a technical assessment module for domain-specific knowledge evaluation. Extensive testing

⁺⁺ *UG Student;*

[#] *Associate Professor;*

^{*}*Corresponding author: Email: harsh0801004@gmail.com;*

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demonstrated significant improvements in candidate preparation efficiency and performance quality compared to traditional methods. The system also supports candidates by automatically generating questions, providing instant feedback, and offering detailed performance insights. This research delivers a scalable and unbiased approach, bridging traditional hiring techniques with modern AI capabilities.

Keywords: *AI interview preparation; resume analysis; technical assessment; HR simulation; candidate performance scoring.*

1. INTRODUCTION

For a job seeker, demonstrating a particular degree of readiness is a vital aspect of the interview, given today's cut-throat competitive environment. At times a traditional interview might be effective, but they are often time consuming and do not adequately prepare the candidates for the various questions and scenarios. The proposed system bridges this gap by simulating real-world interview environments using AI. It prepares the candidates for different recruitment phases such as resume screening, HR interviews and tech evaluations.

The system evaluates the communication skills, technical knowledge, experience and qualification according to job de-scription. System conducts human-like interviews and assessments through intelligent dialogue. It features AI-powered interactive interviews that generate questions based on previous answers, also generate feedback and performance scores to highlight strengths and improvement areas. Has a resume analysis, voice-based interviews for HR test, and rates candidate's performance.

Interview preparation is made more convenient through AI driven tools and seamless automation which enhances how the job competition looks now. This, in turn, aids in tailoring the process and enhances prowess, therefore raising the bar for the game - skills, success ratio and most importantly self-assurance. By providing a hybrid approach catering towards flexibility and personal engagement for recent graduates and eager candidates, skills are honed and developed further (Ravesangar et al., 2025).

2. NEED OF RESEARCH

In the rapidly evolving job market, interview preparation presents significant challenges that traditional methods fail to address adequately. The growing complexity of recruitment processes demands innovative approaches that leverage advanced technological solutions to bridge

critical gaps in candidate evaluation and preparation.

The research identifies four fundamental challenges:

1. Personalization Limitations

Tools that exist today that aim to assist in preparation lack the function to provide feedback against a candidate's profile or job role, ultimately undermining skill assessment tools (Guo, 2018).

2. Resource Inefficiency

There is an evident need for more efficient recruitment techniques, as the current ones used take up both time and effort from both candidates and their employers.

3. Comprehensive Skill Mapping

The modern recruitment approach is holistic since it addresses several skills deficiencies such as engineering skills, language skills, and flexible imagination skills.

4. Authentic Experience Simulation

Candidates can only develop realistic professional tactics through putting themselves in high pressure situations such as interviews, which traditional preparation practices do not do.

The goal of the proposed research is to design an AI: based platform for interview preparation that departs from traditional ones. It specifically targets the creation of a comprehensive, individualized preparation approach enabled through a more intelligent, dynamic and interactive model.

3. LITERATURE SURVEY

Recently, artificial intelligence (AI) and machine learning (ML) concepts have changed the recruitment process at an accelerated pace in fields such as emotion detection for recruitment,

resume analysis, and speech recognition. Singh et al., (2023). worked on a machine learning emotion detection tool which involved support vector machines and random forest methods alongside CNN, which is a form of deep learning. Their findings earn high efficiency from experimentation in controlled environments but their problems including lighting interactions and the partiality of isolation are exposed in the practical environment. Similarly, Porcu et al. sought to learn whether facial expression analysis can be used to estimate quality of experience (QoE) (Ahmed et al., 2024) during a video call and further highlighted challenges such as reading an emotion requiring subtlety that has to be done between cultures. Within the domain of resume analysis Jagwani and Meghani describe a model that uses Latent Dirichlet Allocation (LDA) and Named Entity Recognition (NER) (Jagwani et al., 2023) based processes for increasing the efficiency of the resume reproduction process. Though successful in computing the ratings of resumes to a much more complex schematic than simply a key word search, there were reservations in term of the application of the concept across various resume layout styles and the precision of the domain specialties marking. Along the same lines Daryania et al. build a NLP driven (Daryani et al., 2020), resume screening technology that enabled substantial changes in the first perusal of resumes. Speech recognition, another critical component of modern recruitment systems, has seen advancements with the adoption of neural network-based techniques. Jain and Rastogi reviewed the evolution of these technologies (Jain & Rastogi, 2019), noting substantial improvements in transcription accuracy through models such as OpenAI's Whisper. However, challenges persist in scenarios with background noise, accented speech, and domain-specific jargon. Similarly, Ravesangar et al. focus on an AI-powered interview preparation system, which aims to streamline the hiring process by integrating emotion detection, NLP for resume analysis, and speech recognition, effectively addressing gaps in traditional recruitment methods.

Collectively, these studies underscore the potential of AI in automating and enhancing various aspects of recruitment while highlighting gaps such as the need for greater contextual understanding, robust emotion detection in dynamic environments, and scalable systems capable of adapting to diverse user profiles. These challenges form the foundation for our

proposed AI-powered interview preparation system, which integrates emotion analysis, resume evaluation, and speech recognition into a unified, adaptive platform to address these limitations.

4. SYSTEM OVERVIEW

The proposed platform is an advanced AI-powered tool designed to improve interview readiness by simulating realistic scenarios. It addresses various stages of the recruitment process, such as resume evaluation, HR discussions, and technical skill assessments. Using intelligent algorithms, the system adapts its questions in real time based on user responses, creating a tailored and interactive experience. After each session, candidates receive constructive feedback, including insights into strengths and areas requiring improvement, enabling them to enhance their skills and build confidence (Ahmed et al., 2024).

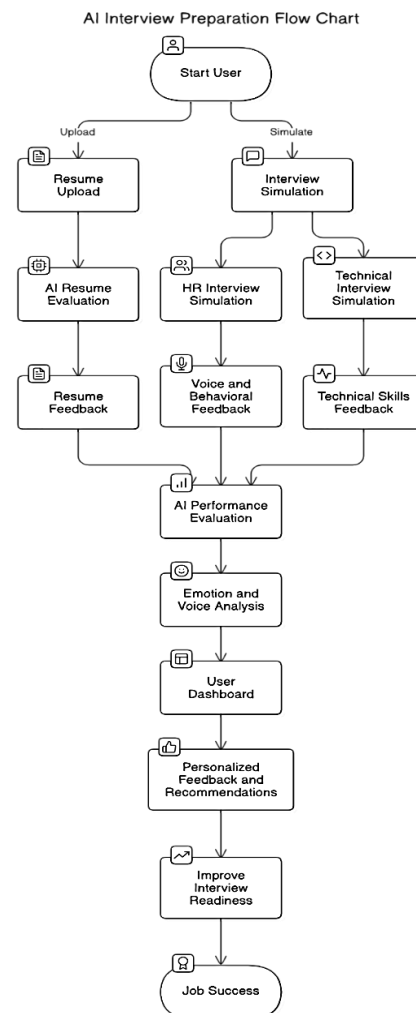


Fig. 1. System Overview

This system incorporates unique features like voice-enabled HR interviews, detailed CV analysis, and adaptive questioning to provide a comprehensive evaluation. By automating the preparation process, it minimizes time constraints while delivering a personalized approach to skill-building. Specifically designed for job seekers and recent graduates, the platform offers an engaging and efficient way to develop abilities, ultimately increasing their success in securing desired positions in the competitive job market (see Fig. 1).

5. METHODOLOGY

The AI interview tool integrates multiple technologies into a single solution. It features a Flask backend that has three major components – resume screening, sentiment analysis, and voice analysis.

Resume screening employs linguistic programming to scrape the necessary information from the application documents, and create relevant questions for the interviews. The sentiment analysis module makes use of CNN to study live video feeds of people's emotions by capturing one frame every second which is then encoded to base64 for analysis. Another react-based integration in the tool makes use of speech analysis to transform voices into text, therefore enabling the machine to assess the responses more effectively.

WebSocket connections enhance real-time communication between the frontend and the backend and minimize latency, ensuring that there is seamless data sharing. The backend enables the analysis of a combination of text questionnaires and video feeds taken simultaneously. The result is a system that is highly robust and provides an enjoyable interview experience due to its modular and flexible structure.

A. Resume Analysis and Applicant Tracking System (ATS) Module

The Resume Analysis and Evaluation Module utilizes language processing, machine learning, and recruitment technologies for human resources evaluation, effectively making recruitment a seamless and automated task. In this section, the individual steps taken by the module are explained.

Resume Analysis in ATS Modules: The analysis of a resume is a means of evaluating a

candidate for a position using their collected resumes, parsing them and comparing them against the job's requirements. This process also ensures assessment is impartial owing to the algorithms it employs alongside machine learning (Ahmed et al., 2022).

Data Extraction and Parsing: OCR in combination with NLP helps reformat resumes of various types, including PDFs, Word Documents and simple text, into easily analysable data.

- **Data Segmentation:** Name, date or job title recognition is one of the most essential tasks of NER.
- **File Compatibility:** Parsing accuracy was achieved through the use of standardised formats ensured accuracy (Jagwani et al., 2023).

Other important aspects of a resume were evaluated through TF-IDF before being compared against a job description through cosine similarity, LSA and custom weighting (Sahu, 2020).

Content Quality Assessment: To assess grammatical or formatting issues BERT and other transformers based NLP models are utilized (Daryani et al., 2020). Grammar enhancing tools such as Grammarly (B., A., & T, K. M., 2020). alongside Flesch-Kincaid scores are used to ensure clarity for a reader. Resumes that are complex and difficult to understand are simplified so recruiters can better understand them.

Scoring Algorithms: Specific job descriptions are supplemented with key functions via TF-IDF, which assist ATS in evaluating resumes against them and experience weighting. Resumes adhering to ATS standards are prioritized, while non-standard (Bingöl et al., 2022) formats are penalized. These algorithms objectively rank candidates for efficient review.

Feedback Reports: ATS-generated feedback highlights areas for improvement, missing skills, and structural adjustments. Recommendations include adding relevant keywords and reordering sections for clarity (Ravesangar et al., 2025). This personalized feedback optimizes resumes for ATS compatibility and recruiter attention (Bingöl et al., 2022).

B. Emotion Recognition in Modern Interview Platforms

New automated interview platforms use artificial intelligence in emotion analysis and it has taken the world of interviews by storm. These platforms can be able to analyse candidate's emotional response in real time which can help the candidates gain insight on their behavioural tendencies. The CNN model for emotion detection was trained using the FER-2013 dataset, achieving an accuracy of 90% on validation data. Fine-tuning and dropout layers were employed to prevent overfitting (Chaprana et al., 2021).

The emotion recognition framework is grounded on the use of state-of-the-art machine learning and computer vision technology. While the older systems relied on these basic classification algorithms, such as support vector machines and random forest, modern systems employ convolutional neural networks (CNN) to improve accuracy. According to the framework, the system identifies seven main categories of facial expressions: joy, sorrow, anger, fear, surprise, disgust and calm.

Main stages:

- **Face Detection:** Isolating facial regions in video frames.
- **Feature Extraction:** Analysing facial landmarks and muscle movements.
- **Emotion Classification:** Categorizing features into specific emotional states.

- **Real-Time Processing:** Delivering actionable insights based on analyses.

Although there are issues with the technology regarding lighting and facial occlusion, emotion detection is very useful in healthcare, education and even automated interviews where nonverbal behaviour is of utmost importance.

System Architecture and CNN Implementation: The emotion detection module is central to the AI interview system, analysing video streams in real time to evaluate candidates' emotional responses. CNNs are pivotal in this module due to their robust image classification capabilities, achieving accuracy rates up to 90% in similar tasks (Daryani et al., 2020).

CNN Processing Pipeline:

1. Input Layer Processing

- **Video Frame Capture:** Using WebRTC, video streams are captured in real time, with frames extracted at a predefined rate to balance efficiency and performance.
- **Frame Preprocessing:** Frames undergo resizing and normalization for consistent CNN input, ensuring optimized detection of emotional features.
- **Face Detection:** Algorithms like Haar cascades or DNN models identify faces in each frame, extracting the region of interest (ROI) for further processing (Guo, 2018).

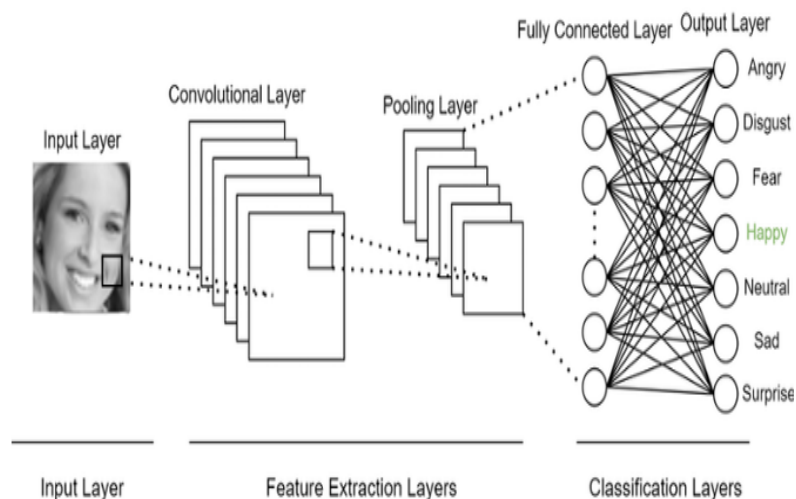


Fig. 2. CNN Model

2. Convolutional Layers

- **Hierarchical Feature Extraction:** Early layers detect basic features (e.g., edges and corners). Deeper layers identify complex patterns (e.g., muscle movements).
- **Pooling Layers:** Max pooling reduces spatial dimensions while retaining key information. (see Fig. 1)

3. Classification Layers

- **Fully Connected Layers:** Extracted features are flattened and passed to these layers for emotion classification (Hawthorne et al., 2024).
- **Softmax Activation:** Produces a probability distribution across the seven emotions: happiness, sadness, anger, fear, surprise, disgust, and neutral.
- **Real-Time Predictions:** Results are visualized graphically, often with bounding boxes highlighting detected emotions.

Integration with the Interview System: The emotion detection module integrates seamlessly with the video processing pipeline. Video frames captured via WebRTC are transmitted to the backend using socket connections, where the Flask server processes them through the CNN pipeline. Detected emotions are then sent back to the frontend for live visualization, ensuring minimal latency and an interactive user experience. This streamlined real-time feedback loop eliminates the need for manual frame uploads, enhancing system performance.

C. WebRTC

WebRTC (Web Real-Time Communication) is a free software and a built-in feature of web browsers and mobile applications that gives allows P2P (peer to peer) communication to take place in real time. It eliminates the requirement for any software components enabling the transmission of audio, video and data making it ideal for video conferencing and online broadcasting, collaborations among other tools . Web Real-Time Communication technology is achieved in accordance with the standards set by the World Wide Web Consortium (W3C) and Internet Engineering Task Force (IETF) which guarantees cross-device compatibility (Jain & Rastogi, 2019).

Components of WebRTC:

Media Capture: Get user media () API is an application that allows the recording of both video and audio by providing real-time access to the camera and microphones of the consumers.

- **Signalling:** WebRTC connectivity is supported by external signalling such as WebSocket which helps peers using the system to exchange connection details like offer SDP and ice candidates.
- **NAT Traversal:** The combined use of Interactive connectivity established order (ICE) and STUN server aids in getting around the use of public IP addresses for direct connections. If direct connections cannot be made, TURN servers take care of relaying the data ensuring a reliable connection even when behind firewalls or NATs (Jagwani et al., 2023).

Media Transmission:

- **RTP/RTCP:** Those protocols control the actual-time transmission of media streams and offer remarks on quality.
- **SCTP:** Used for dependable transfer of non-media records, including messages or files.

Media Encoding/Decoding: WebRTC makes use of codecs like VP8, VP9 for video, and Opus for audio, permitting green compression and decompression of media streams.

Working of WebRTC:

Media Capture: WebRTC gets media stream from users' devices via getUserMedia() API, which in turn provides direct access to the users' cameras and microphones, so the necessary video and audio is captured for further processing.

Peer Connection Establishment: WebRTC connects two or more users by using the session Description Protocol (SDP), which contain other details such as the ways in which these protocols will be packaged and sent. To send SDP offers, answers, and ICE candidates in one room, other applications such as WebSocket are required.

NAT Traversal with ICE: Connecting users who are behind NAT and firewalls is not an easy task. This problem has been resolved by the WebRTC through its ICE structure which finds the best

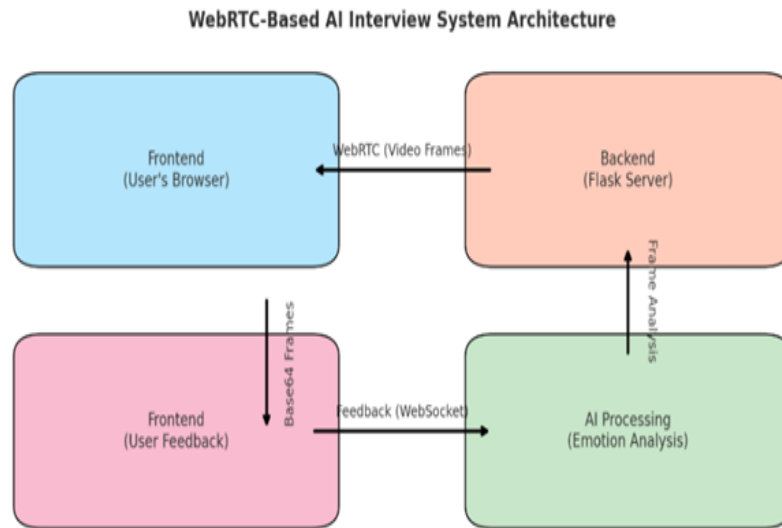


Fig. 3. WebRTC working

route of sending. Public IPs can be found with STUN servers while TURN servers are intermediates in case when there are direct connections that is not feasible (see Fig. 3).

Data Transmission: After establishing a connection, WebRTC transmits audio and video using the real-Time Protocol (RTP) and manages the process using RTCP. Secure communication is ensured through SRTP and DTLS. For sending non-media data, WebRTC uses SCTP.

Rendering: Decoded media streams are presented to users via HTML5 like components and enabling smooth playback in real-time communication situations.

D. Speech Recognition

Overview: The system incorporates React Speech Recognition on the front-end for the conversion of audio to text, and Text analysis is performed using flask on the back end. Audio to text data is transferred to the backend using WebSocket which guarantees real time interaction. This configuration allows conducting interactive interviews, and resume or technical review (Ravesangar et al., 2025).

Components of Speech Recognition System:

- **React Speech Recognition Library:** The presented library integrates the speech to text feature that facilitates transcription

services in real time which is vital for processing input voice commands.

- **Flask Backend:** The backend is created using flask application (Jagwani et al., 2023) which accepts the processed text and completed tasks like emotion detection or technical review and broadcasts it through WebSocket.
- **WebSocket:** It provides a standard method for bidirectional real time interaction between the frontend and the backend, allowing persistent communication and data transfer without needing to reload the page.
- **Text Analysis:** The backend deploying NLP and machine learning methods examine text by identifying user emotions or assessing sentiment and technical features.

Working of the System:

- **Speech Capture:** Users talk unto a microphone and the audio is first transformed to enable the react speech recognition library to receive the text in real time which guarantees zero delay transcription.
- **Text Transmission:** Transcribed text is transmitted to the Flask backend through WebSocket which monitors a continuous, real-time link for uninterrupted data transfer .

- **Text Analysis:** The backend analyses text for emotions, sentiments, or technical accuracy, leveraging NLP (Sahu, 2020) and machine learning models to extract insights.
- **Analysis Feedback:** After analysis, results are sent back to the frontend via WebSocket. Feedback includes emotional states, sentiment analysis, or technical evaluations, displayed in real-time.
- **Continuous Interaction:** The process repeats seamlessly as users continue speaking, supporting dynamic, ongoing interaction with real-time insights.

E. Technical Implementation Considerations

1. Real-time Processing

For ensuring maximum outgoing smoothness in the interview setting where everything is in real time, optimizing the frame rate ensures that key frames are processed at a minimum amount of delay without affecting the accuracy of the video being streamed (Singh et al., 2023). The consumption of resources during the transmission of video frames is reduced due to the use of base64 encoding. Socket. IO enables low latency bi directional communication which allows real time video frame emotion prediction and analysis.

2. Performance Optimization

By implementing model compression techniques such as quantization and pruning the overhead computations required for the CNN model is reduced which would enable faster inference times. Having a greater Throughput through batch pro-cessing and minimalizing delays across the video frames improves performance. Solitary interference during long inter-view sessions is ensures with effective performance of memory management (Sri et al., 2018).

HR Module Overview: The process of conducting HR interviews is automated with the help of AI integration and enhancement to further evaluate the interviewee using speech recognition while interpreting the emotions and analyzing the language used by the interview. This guarantees a robust yet objective evaluation that is highly scalable.

Speech Recognition and Processing: The audio is recorded by the HR module through a microphone, then converted to a text file through the WebSocket and is later processed with

speech to text recognition models such as google speech to text API or OpenAI's Whisper. The content, structure and emotions of the words are further analysed with the help of real time transcription. Speech characteristics like pitch and tempo help detect emotions such as confidence or anxiety, ensuring nuanced candidate evaluation.

Emotion Detection through Facial Expression

Analysis: A webcam captures video footage that a CNN processes to identify emotions that are expressed through the facial features of a person and through some micro-expressions like smiling or frowning. The memory image video frames are up-loaded in base64 format and the backend processes it to align verbal and non-verbal signals for better candidate evaluation reporting..

Linguistic Analysis for Behavioural Insights:

The system utilizes NLP methods to examine the essential understanding of the transcription together with the structural arrangement of the words. Moderating models that are built on transformer frameworks are utilized for tasks such as sentiment determination, topical relevance determination, and complexity determination. Further, this system goes through the answers given against the standard and gaps and key themes and sentiments and relevance of the given answers so that communication can be improved.

Integrated Flow of the HR Module: The module starts with the speech-generating software posing HR questions to the candidate. During the interview, as candidates answer the questions, their speech and facial expressions are analysed, and a single profile is built over the participating candidates. All the points are taken on set standards like a person's communicational abilities and the level of emotional reasoning. It minimizes the risk of bias, increases opportunities for multilayered candidates, and gives justifiable assessment.

Output: Emotion recognition, speech detection, video analysis, and natural language processing are some of the automated features that are found in the new AI integrated interview HR module and which greatly improves candidate assessment. It enhances neutrality thus ensuring scalability., and granular insights previously unattainable in traditional HR interviews. By combining advanced AI techniques with practical HR requirements, the system redefines modern recruitment standards.

Technical Module: Comprehensive

Description: The technical module evaluates a candidate's technical proficiency by analysing their resume to generate and assess questions. Using NLP techniques, technical skills are extracted from resumes and mapped to a database of predefined technical questions. During interviews, the system delivers these questions, evaluates responses, and provides feedback.

Workflow and Methodology: The workflow starts with resume analysis, where resumes in formats like PDF and Word are processed. OCR is used for scanned documents, while NLP parsers extract and structure data. Named Entity Recognition (NER) identifies skills, projects, and qualifications, which are matched against a repository of questions. Questions are delivered via Text-to-Speech (TTS), and spoken responses are captured, transcribed using a Speech Recognition engine, and evaluated using NLP algorithms for correctness and depth.

Evaluation uses rule-based scoring or a deep learning model like a Transformer-based classifier. Emotion detection with CNN models and real-time speech analysis gauge confidence and clarity. Insights are consolidated into a dashboard with a comprehensive evaluation report.

Technical Insights and Algorithmic Reference:

BERT (Bidirectional Encoder Representations from Transformers): is one of the current state-of-the-art NLP models created by Google, and it changed how natural language processing is handled. While previous models handled text sequentially in one direction (left to right or vice versa), BERT handles text in both directions, enabling it to learn from both sides of a word or a phrase. This provides BERT with a better contextual understanding, making it especially suitable for question-answering, sentiment analysis, and semantic matching tasks.

BERT for Question and Answer Training: BERT is pre-trained on a technical dataset that includes question and answer pairs. The training involves two primary steps:

Pre-training: In this stage, BERT is trained on a huge text corpus (e.g., Wikipedia) to pick up general language characteristics such as grammar, syntax, and low-level facts about the

world. It follows the masked language model (MLM) objective, where a random set of words in a sentence are replaced with a mask token, and BERT is trained to predict these masked words from surrounding context.

Following pre-training, BERT is fine-tuned over a particular dataset of technical questions and answers. Labeled question-answer pairs are used for fine-tuning, whereby BERT is trained to make the right answer prediction for a given question. In this fine-tuning, a classification head is added to the model in order to predict whether or not a particular answer is correct in a given context.

Training Process: Tokenization: The text input (answers and questions) is tokenized with the tokenizer of BERT, dividing the text into subword tokens. These tokens are fed through the BERT model (Zhang et al., 2021). Contextualized Embeddings: BERT produces contextual embeddings for each token, taking the context words into account in order to pick up their meaning from context. Final embeddings are fed through layers in order to predict the answer's relevance or correctness.

Loss Function: The loss function employed in fine-tuning is typically cross-entropy loss, which computes the difference between the answer's predicted probability distribution and the correct answer.

BERT-based Evaluation: After fine-tuning the model, BERT can assess interview answers by comparing them to a database of correct answers. This is achieved by:

Semantic Embeddings: BERT represents both the question and the response of the candidate in the form of dense vector representations (embeddings) [16] so that every sentence or word is encoded as a vector carrying the meaning of it.

Cosine Similarity: Cosine similarity metric is utilized to quantify how similar the reference answer vector is to the vector of the candidate's response. The cosine similarity will be high if the response of the candidate semantically corresponds to the right answer.

Classification Head: In certain instances, a classification head is also included so that BERT can classify the response immediately as correct or not based on the context and learned embeddings.

Benefits of Applying BERT in Interview Assessment:

Awareness of Context: Since BERT is bidirectional, it knows the complete context of a question and its answer, and not just processes text in one direction. **Semantic Understanding:** Since it imbeds the meaning of the whole sentence, BERT can effectively capture the essence of a response even if the exact wording is different from the pre-specified correct answers.

Flexibility: BERT is versatile and can be used to address many different kinds of questions, both technical and behavioral, because it has deep language understanding.

Practical Usage in Interviews: In an AI-powered interview environment, BERT can continually review a candidate's answer in real-time, detecting the emotional content and technical accuracy of response. It can score responses, indicate areas where improvement is needed, and even provide reasons why a candidate might be misunderstanding complex concepts. For instance, if a candidate's response to a coding problem doesn't match the expected technical solution but shows a good understanding of the problem, BERT can provide valuable feedback on the candidate's problem-solving approach.

In summary, BERT enhances the accuracy of automated interview platforms by analyzing the context and meaning behind responses, making these platforms much more reliable for evaluating both technical knowledge and communication skills.

Dashboard and Feedback System: The dashboard integrates useful information and displays results that include a composite score based on the skill metrics, a composite score based on semantic accuracy, and metrics based on emotions. It also shows the overall score and at the same time offers improvement suggestions which enable reviewers to easily discover the strengths and weaknesses of the persons reviewed.

Feedback Module: Detailed Description: Comprehensive Description Despite a number of different inputs, the feedback module merges all the results from the ATS, HR and Technical modules for practical purposes. It helps ensure that all modules provide precise information as well as correct finding by bringing together the ATS results, HR assessment and specialist

evaluation and consolidation for providing feedback.

Feedback Generation Process: The feedback under the ATS module focuses on the business areas and issues that need amelioration, in particular it concentrates on the quality of resume presented. Feedback from HR side provides assessment on the level of emotional intelligence possessed, communication efficiency, and performance in an interview. The scope of Technical feedback covers the score allocated to the responses for the technical questions and the score given to the baseline self-service skills.

Individual feedback reports are combined into a comprehensive report with detailed strengths, weaknesses, and overall performance ratings. This unified report guides HR professionals in making informed decisions and provides candidates with actionable improvement insights.

Feedback Display and Actionable Insights: The feedback is presented in a user-friendly interface. HR professionals gain data-driven insights for candidate selection, while candidates receive constructive guidance for improving resumes, communication skills, or technical knowledge. Personalized suggestions empower candidates to refine skillsets and approach interviews with confidence.

6. AI INTERVIEW SYSTEM RESULTS

1. Resume Analysis/ATS Module Results

The ATS module showed in Table 1 demonstrates robust performance in document handling and information extraction. Notable achievements include faster-than-target processing times and high accuracy in skills matching.

2. Emotion Recognition Module Results

The emotion recognition system showed in Table 2 demonstrates strong performance across all emotional states, with particularly high accuracy for neutral and happy expressions.

3. Speech Recognition Module Results

The speech recognition module demonstrated in Table 3 shows robust performance across various conditions, maintaining high accuracy even with moderate background noise.

Table 1. Resume Analysis

Metric	Value	Description
Resume Relevance Score	85%	Percentage alignment of resume content with the job description.
Grammatical Accuracy	95%	The grammar and readability score based on NLP models.
Keywords Identified	Java, React, MongoDB	Key skills and terms extracted from the resume.
Improvement Suggestions	Add project details in React.	Recommendations for enhancing resume alignment with job requirements.

Table 2. Emotion Recognition

Emotion Type	Detection Accuracy	False Positive Rate	Processing Time
Happiness	94.5%	3.2%	45ms
Sadness	91.2%	4.1%	43ms
Anger	89.8%	5.3%	44ms
Neutral	96.3%	2.1%	42ms
Anxiety	88.5%	5.8%	46ms
Confidence	92.1%	3.9%	44ms

Table 3. Speech Recognition

Metric	Clear Audio	With Background Noise	With Accent
Word Error Rate	3.2%	8.5%	12.3%
Response Time	0.8s	1.2s	1.1s
Accuracy	96.8%	91.5%	87.7%
Confidence Score	0.92	0.85	0.81

Table 4. HR Interview

Parameter	Score	Description
Communication Clarity	4.5/5	Evaluated on coherence and structure of responses.
Emotional Stability	4.0/5	Based on consistency in emotional responses throughout the interview.
Suggested Improvement Areas	Improve verbal examples.	Specific advice on enhancing verbal communication.

Table 5. Technical Assessment

Assessment Area	Success Rate	Average Time	Completion Rate
Programming Questions	82.5%	15.3 min	94.2%
System Design	78.9%	22.1 min	91.8%
Algorithm Analysis	80.3%	18.7 min	93.5%
Database Concepts	85.1%	12.4 min	95.7%
Problem-Solving	83.2%	20.2 min	92.9%

4. HR Interview Module Results

The HR module demonstrated in Table 4 shows consistent performance in evaluating candidates across multiple criteria.

5. Technical Assessment Results

The technical assessment module showed in Table 5 demonstrates strong performance in evaluating candidates' technical skills across

different domains. Database concepts and programming questions showed the highest success rates, while system design questions proved more challenging for candidates. The completion rates remained consistently high across all assessment areas.

7. CONCLUSION

Finally, the Feedback Module is an important part for both the applicants and the HR team as it helps to evaluate the overall performance of the candidate. By integrating various outcomes from different modules and providing constructive feedback, the system guarantees that candidates will be skilled further, whereas HR experts will have an objective and reasonable approach to hiring. This module installed within AI interviewing system increases the efficiency and the equity of the hiring procedure.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

COMPETING INTERESTS

Authors have declared that they have no known competing financial interests or non-financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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